Chapter 12, Problem 23P

Problem

A balanced delta connected source is connected to a balanced delta connected load where *ZL* = (80 + *j*60) Ω and *Zl* = (2 + *j*) Ω. Given that the load voltages are

$\mathbf{V}_{AB} = 100\angle 0^\circ$ V, $\mathbf{V}_{BC} = 100\angle 120^\circ$ V, and

$\mathbf{V}_{CA} = 100\angle -120^\circ$ V. Calculate the source voltages **V***ab*, **V***bc*, and **V***ca*.

Step-by-step solution

Step 1 of 10

Consider the circuit diagram shown in Figure 1.

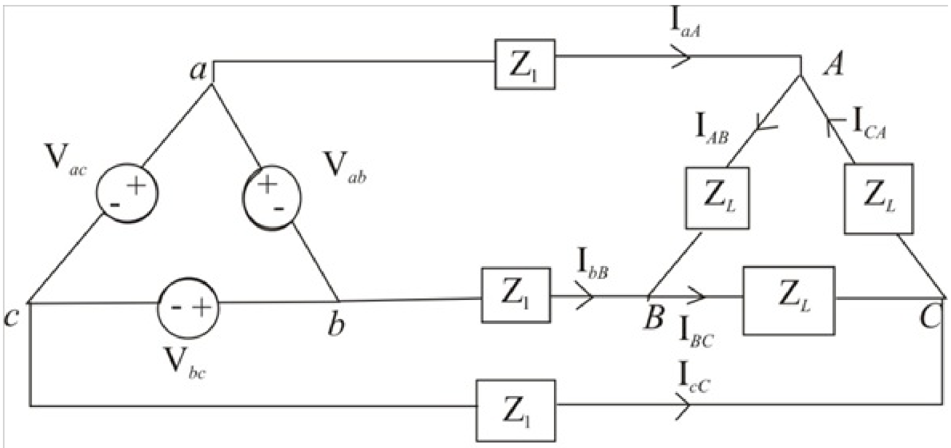

Figure 1

Step 2 of 10

Calculate the value of the current I_{AB}.

$$I_{AB} = \frac{V_{AB}}{Z_L}$$

Substitute $100\angle 0^\circ$ V for V_{AB} and $80 + j60\ \Omega$ for Z_L.

$$\begin{aligned} I_{AB} &= \frac{100\angle 0^\circ}{80 + j60} \\ &= \frac{100\angle 0^\circ}{100\angle 36.87^\circ} \\ &= 1\angle -36.87^\circ \text{ A} \end{aligned}$$

Therefore, the value of the current I_{AB} is, $1\angle -36.87^\circ$ A.

Step 3 of 10

Calculate the value of the current I_{BC}.

$$I_{BC} = \frac{V_{BC}}{Z_L}$$

Substitute $100\angle 120°\ V$ for V_{BC} and $80 + j60\ \Omega$ for Z_L.

$$\begin{aligned} I_{BC} &= \frac{100\angle 120°}{80 + j60} \\ &= \frac{100\angle 120°}{100\angle 36.87°} \\ &= 1\angle 83.13°\ A \end{aligned}$$

Therefore, the value of the current I_{BC} is, $1\angle 83.13°\ A$.

Step 4 of 10

Calculate the value of the current I_{AC}.

$$I_{AC} = \frac{V_{AC}}{Z_L}$$

Substitute $100\angle -120°\ V$ for V_{AC} and $80 + j60\ \Omega$ for Z_L.

$$\begin{aligned} I_{AC} &= \frac{100\angle -120°}{80 + j60} \\ &= \frac{100\angle -120°}{100\angle 36.87°} \\ &= 1\angle -156.87°\ A \end{aligned}$$

Therefore, the value of the current I_{AC} is, $1\angle -156.87°\ A$.

Step 5 of 10

Calculate the value of the line current I_{aA}.

$$I_{aA} = I_{AB} - I_{AC}$$

Substitute $1\angle -36.87°\ A$ for I_{AB} and $1\angle -156.87°\ A$ for I_{AC}.

$$\begin{aligned} I_{aA} &= 1\angle -36.87° - 1\angle -156.87° \\ &= 1.732\angle -6.87°\ A \end{aligned}$$

Therefore, the value of the line current I_{aA} is, $1.732\angle -6.87°\ A$.

Step 6 of 10

Calculate the value of the line current I_{bB}.

$$I_{bB} = I_{BC} - I_{AB}$$

Substitute $1\angle -36.87°\ A$ for I_{AB} and $1\angle 83.13°\ A$ for I_{BC}.

$$\begin{aligned} I_{bB} &= 1\angle 83.13° - 1\angle -36.87° \\ &= 1.732\angle 113.13°\ A \end{aligned}$$

Therefore, the value of the line current I_{bB} is, $1.732\angle 113.13°\ A$.

Step 7 of 10

Calculate the value of the line current I_{cC}.

$$I_{cC} = I_{CA} - I_{BC}$$

Substitute $1\angle -156.87^\circ$ A for I_{CA} and $1\angle 83.13^\circ$ A for I_{BC}.

$$\begin{aligned} I_{cC} &= 1\angle -156.87^\circ - 1\angle 83.13^\circ \\ &= 1.732\angle -126.87^\circ \text{ A} \end{aligned}$$

Therefore, the value of the line current I_{cC} is, $1.732\angle -126.87^\circ$ A.

Step 8 of 10

Calculate the value of the source voltage V_{ab}.

$$\begin{aligned} V_{ab} &= I_{aA}Z_L + V_{AB} - I_{bB}Z_L \\ &= \left(I_{aA} - I_{bB}\right)Z_L + V_{AB} \end{aligned}$$

Substitute $1.732\angle -6.87^\circ$ A for I_{aA}, $1.732\angle 113.13^\circ$ A for I_{bB}, $100\angle 0^\circ$ V for V_{AB} and $2 + j1\ \Omega$ for Z_l.

$$\begin{aligned} V_{ab} &= \left(1.732\angle -6.87^\circ - 1.732\angle 113.13^\circ\right)\left(2 + j1\right) + 100\angle 0^\circ \\ &= \left(3\angle -36.87^\circ\right)\left(2.236\angle 26.57^\circ\right) + 100\angle 0^\circ \\ &= 6.708\angle -10.3^\circ + 100\angle 0^\circ \\ &= 106.61\angle -0.65^\circ \text{ V} \end{aligned}$$

Therefore, the value of the line voltage V_{ab} is $\boxed{106.61\angle -0.65^\circ \text{ V}}$.

Step 9 of 10

Calculate the value of the source voltage V_{bc}.

$$\begin{aligned} V_{bc} &= I_{bB}Z_L + V_{BC} - I_{cC}Z_L \\ &= \left(I_{bB} - I_{cC}\right)Z_L + V_{BC} \end{aligned}$$

Substitute $1.732\angle -126.87^\circ$ A for I_{cC}, $1.732\angle 113.13^\circ$ A for I_{bB}, $100\angle 120^\circ$ V for V_{BC} and $2 + j1\ \Omega$ for Z_l.

$$\begin{aligned} V_{bc} &= \left(1.732\angle 113.13^\circ - 1.732\angle -126.87^\circ\right)\left(2 + j1\right) + 100\angle 120^\circ \\ &= \left(3\angle 83.13^\circ\right)\left(2.236\angle 26.57^\circ\right) + 100\angle 120^\circ \\ &= 6.708\angle 10.97^\circ + 100\angle 120^\circ \\ &= 106.61\angle 119.36^\circ \text{ V} \end{aligned}$$

Therefore, the value of the line voltage V_{bc} is $\boxed{106.61\angle 119.36^\circ \text{ V}}$.

Step 10 of 10

Calculate the value of the source voltage V_{ca}.

$$\begin{aligned} V_{ca} &= I_{cC}Z_L + V_{CA} - I_{aA}Z_L \\ &= \left(I_{cC} - I_{aA}\right)Z_L + V_{CA} \end{aligned}$$

Substitute $1.732\angle -126.87^\circ$ A for I_{cC}, $1.732\angle -6.87^\circ$ A for I_{aA}, $100\angle -120^\circ$ V for V_{CA} and $2 + j1\ \Omega$ for Z_l.

$$\begin{aligned} V_{ca} &= \left(1.732\angle -126.87^\circ - 1.732\angle -6.87^\circ\right)\left(2 + j1\right) + 100\angle -120^\circ \\ &= \left(3\angle -156.87^\circ\right)\left(2.236\angle 26.57^\circ\right) + 100\angle -120^\circ \\ &= 6.708\angle -130.3^\circ + 100\angle -120^\circ \\ &= 106.61\angle -120.65^\circ \text{ V} \end{aligned}$$

Therefore, the value of the line voltage V_{ca} is $\boxed{106.61\angle -120.65^\circ \text{ V}}$.